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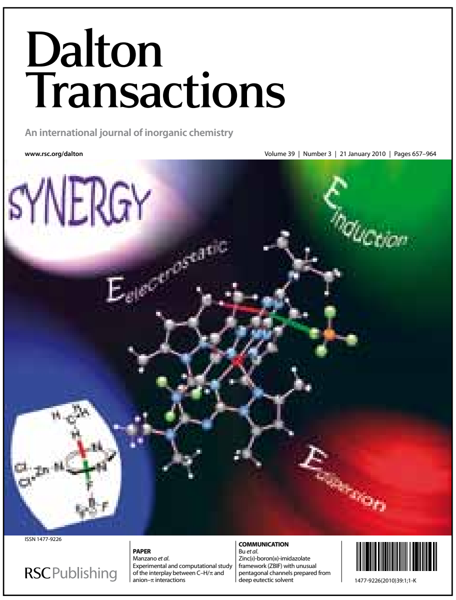

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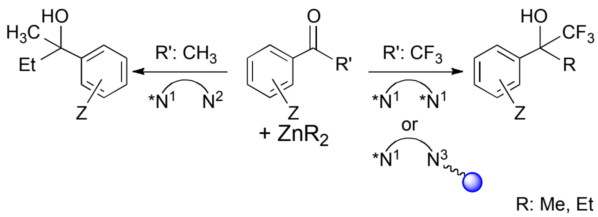
A smart molecular design of the catalyst allows for a chemo- and enantioselective method for the preparation of chiral fluorinated or non-fluorinated tertiary alcohols.



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ARTICLE TYPE

Molecular and Merrifield supported chiral diamines for enantioselective addition of ZnR_2 (R = Me, Et) to ketones

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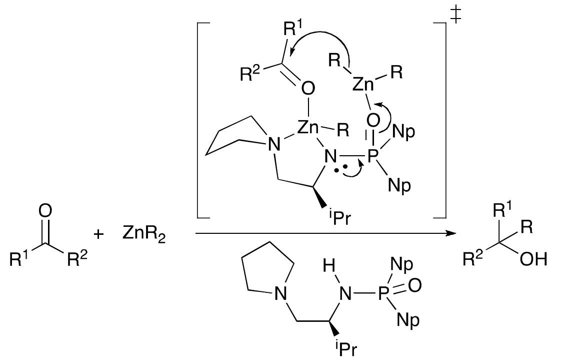
Chiral 1,2-ethylenediamines have been previously reported as active catalysts in the enantioselective addition reactions of ZnR_2 to either methyl- or trifluoromethyl- ketones. Subtle changes in the molecular structure of different catalysts are described herein and lead to a dramatic effect in their catalytic activity. From these findings, we demonstrate the selective reactivity of the ligands used in the addition of ZnR_2 (R = Me, Et) to methyl- and trifluoromethyl-ketones offering an enantioselective access either to chiral non fluorinated alcohols or to chiral fluorinated tertiary alcohols. Considering the importance of the chiral trifluoromethyl carbinol fragment in several biological active compounds, we have extended the scope of the addition reaction of $ZnEt_2$ to several trifluoromethylketones catalyzed by (R,R)-1,2-diphenylethylenediamine derivatives. This work explores a homogeneous approach that provides excellent yields and very high ee and the use of a heterogenized tail-tied ligand affording moderate ee, high yields and allowing an easier handling and recycling.

Introduction

Asymmetric organozinc additions to prochiral carbonyl compounds allow for the synthesis of chiral alcohols that are present in the structure of many natural products and synthetic pharmaceuticals.[1] These reactions have been the subject of exhaustive research and are dealt with in a number of reviews.[2] Trifluoromethyl ketones are particularly interesting because they can be used in the asymmetric synthesis of chiral secondary and tertiary trifluoromethylalcohols. The latter provide fragments present in chiral reagents,[3] and in various biologically active compounds behaving as anticancer drugs,[4] anticonvulsants,[5] protease inhibitors,[6] 11*β*-hydroxysteroid dehydrogenase type I (11β-HSD1) enzyme inhibitors,[7] anti-HIV agents,[8,9] etc. The considerable demand for enantiomerically enriched fluorine-containing compounds is attracting scientific activity to the development of new catalytic enantioselective methods for their preparation.[10,11]

Despite all this effort, the asymmetric addition to ketones is still challenging, mainly because of the low reactivity of ketones and the difficulty of controlling the enantiofacial stereoselectivity.[12] Moreover, the organozinc reagents used for this addition can cause the enolization of ketones bearing α-hydrogens , and promote undesired self-aldol reactions.[13] Ketones lacking α-H, such as trifluoromethyl ketones, cannot undergo this undesired side reaction, but they are prone to suffer a reduction process via β-H transfer to give secondary alcohols.[14]

Tailoring an efficient catalyst requires to take into account the electronic and steric features of the substrates, in order to overcome or minimize the possible side reactions, but this task is



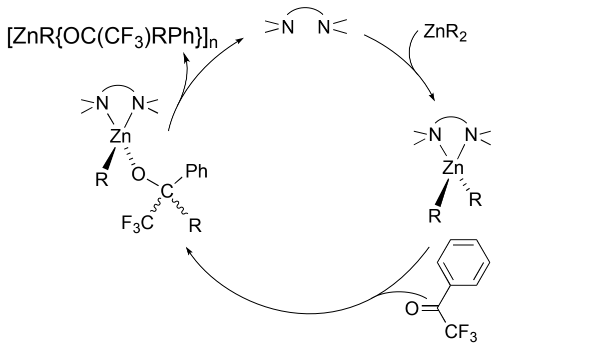
Scheme 1. Catalytic alkylation of R^1R^2CO using Ishihara's catalyst

extremely difficult as different mechanisms seem to be operating.

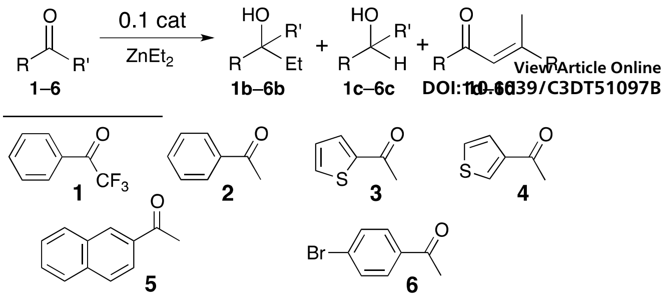
Thus, Ishihara and coworkers have reported a highly efficient organozinc addition to alkyl ketones catalyzed by chiral phosphoramide-Zn(II) complexes.[15,12f] The proposed transition state involves two molecules of ZnR_2 and the formation of a Lewis acid/Lewis base dyad (Scheme 1).

Recently, Hevia and coworkers have proposed a very different mechanism (Scheme 2) for the reaction of $PhCOCF_3$ with $[ZnR_2(TMEDA)]$, based on some experimental evidence detecting the formation of the corresponding addition or reduction products.[16]

Yearick and Wolf screened experimentally a large number of achiral and chiral ligands, including chiral diamines and bisoxazolines, for the addition of $ZnEt_2$ to $RCOCF_3$ ketones.[17] In



Scheme 2. Catalytic cycle for alkylation of $PhCOCF_3$ using TMEDA as catalyst



Scheme 3

Table 1. Results of the addition of $ZnEt_2$ to the aromatic ketones in Scheme 3 with $\mathbf{L^2}$ and $\mathbf{L^1}$.

entry	ket	L	addition **nb** (ee%, conf.)	reduction **nc**[a]	self-aldol **nd**[b]	starting material **n**
1	**1**	$\mathbf{L^2}$	-	100	-	-
2	**1**	$\mathbf{L^1}$	97 (92)	-	-	3
3	**2**	$\mathbf{L^2}$	52 (87,R)[c]	-	45	3
4	**2**	$\mathbf{L^1}$	3	14	62	21
5	**2**	-	-	69	21	10
6	**3**	$\mathbf{L^2}$	19 (91)	45	30	6
7	**4**	$\mathbf{L^2}$	56 (92)	25	18	1
8	**5**	$\mathbf{L^2}$	58	26	15	1
9	**6**	$\mathbf{L^2}$	27 (93)	22	46	5

[a] Racemic. [b] The self-aldol products are in fact a mixture of the α,β-unsaturated ketone (d) and the hydrate derivative. [c] The absolute stereochemistry for 2b was determine by comparing with literature data[12f, 19]

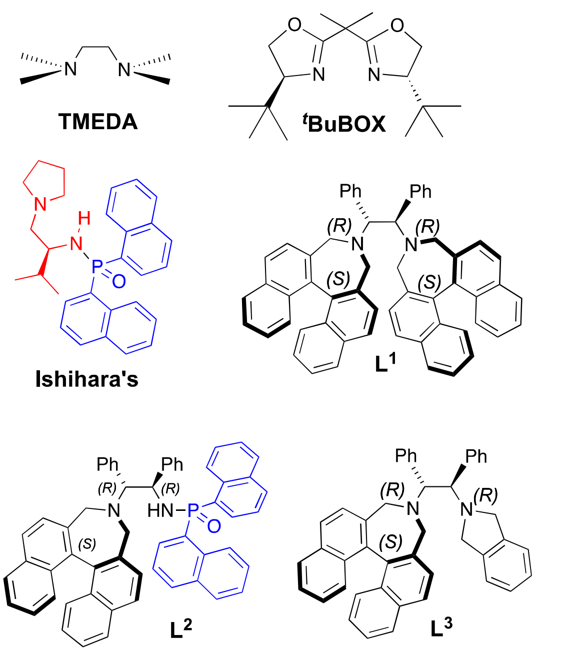


Chart 1. Molecular N–N catalysts

the presence of ^{t}BuBOX (Chart 1) the reactions are relatively fast, but they show much dependence on the R group (71–99% yield with 0–63% ee). TMEDA (93% yield) and ^{t}BuBOX (85% yield with 63% ee at –78 ºC) were the best ligands for the addition of $ZnEt_2$ to several $ArCOCF_3$ ketones. We later reported the enantioselective addition of $ZnMe_2$ and $ZnEt_2$ to 2,2,2-trifluoroacetophenones using the chiral diamines L^1 and L^3, depicted in Chart 1,[18] and found the first successful asymmetric additions of the less reactive $ZnMe_2$ to a variety of trifluoromethyl ketones, giving the corresponding 2-aryl-1,1,1-trifluoropropan-2-ols in high yields, with moderate to high ee (up to 83% for $PhCOCF_3$). With $ZnEt_2$ only the addition to $PhCOCF_3$ was studied, achieving a new record combination of excellent yield (up to 99%) with high enantioselectivity (up to 92%) with L^1, and just somewhat less good results with L^3.

Herein, we report the synthesis and/or the use of different types of catalysts ($\mathbf{L^1}$, $\mathbf{L^2}$, $\mathbf{L^3}$, Ishihara's), or variations of them, beyond their so far examined scope. We have extended the use of ligand $\mathbf{L^1}$ to the addition of dimethyl- and diethylzinc to several commercially available trifluoroacetophenone derivatives containing electron-withdrawing (EWG) and -donating (EDG) groups. Moreover, we have produced a recyclable immobilized derivative of $\mathbf{L^3}$ for the synthesis of trifluoromethylethylalcohols.[20]

Results and discussion

A novel catalyst ($\mathbf{L^2}$) combining features of $\mathbf{L^1}$ and the Ishihara's catalyst (Chart 1) was synthesized, aimed at examining the features that determine the alkylation efficiency. Analogous to Ishihara catalyst, $\mathbf{L^2}$ displays a N-H group for coordination to ZnEt, and a P=O group to act as a Lewis base; and, looking at our previous good results with trifluoromethylketones, $\mathbf{L^2}$ bears the bulky N-substituent of $\mathbf{L^1}$ and $\mathbf{L^3}$. This "hybrid" ligand ($\mathbf{L^2}$) might be able to follow any of the two aforementioned mechanisms for the alkylation of methyl and trifluoromethylketones. The performance of $\mathbf{L^2}$ was tested in the addition reaction of $ZnEt_2$ to a selected library of ketones (Scheme 3), and compared with $\mathbf{L^1}$. The results are summarized in Table 1. Along with the desired addition product (**b**), the secondary alcohol (**c**) and, for methylketones, the self-aldol product (**d**) can be formed as side-products. The working model proposed by Ishihara (Scheme 1), which accounts for the transition-state assemblies in the catalytic enantioselective ethylation, also explains our results, although the bulkier binaphthyl group used in our ligand $\mathbf{L^2}$ noticeably decreases the addition reaction rate (favouring the presence of side products from competitive reactions).

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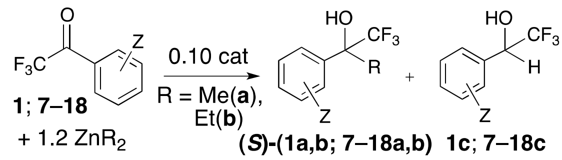

Z = H (**1**), 4-F (**7**), 4-Cl (**8**), 4-Br (**9**), 4-Me (**10**), 4-Et (**11**), 4-^{i}Pr (**12**), 3,5-Me (**13**), 4-SMe (**14**), 4-SEt (**15**), 4-OMe (**16**), 2-OMe (**17**), 4-CO_2Me (**18**)

Scheme 4. Ketones and Zn alkyls used for Table 1. The absolute stereochemistry for **1b, 9a, 9b** was determined by comparing with literature data[21] and the stereochemistry of other alkyl-trifluoromethylated alcohols was assumed by analogy.

In the case of $PhCOCF_3$, $\mathbf{L^2}$ shows no catalytic activity for addition (100% reduction), in contrast with $\mathbf{L^1}$, which provides the addition product with excellent yield and high ee (entries 1 *vs.* 2).[14] Strikingly, $\mathbf{L^2}$ does show moderate to good catalytic activity for addition of $ZnEt_2$ to $ArCOCH_3$ ketones (entries 3-10) and, without further optimization, it provides excellent ee's, but moderate to low yields. On the contrary, $\mathbf{L^1}$ does not catalyze the addition of $ZnEt_2$ to methylketones (entries 3 *vs.* 4). Thus the different catalytic behavior of $\mathbf{L^1}$ and $\mathbf{L^2}$ offers enantioselective access either to chiral fluorinated tertiary alcohols (with $\mathbf{L^1}$) or to chiral non fluorinated alcohols (with $\mathbf{L^2}$).[14] The selectivity of $\mathbf{L^1}$ towards the addition of $ZnEt_2$ to trifluoromethylketone was proven in the case of 1-(4-acetylphenyl)-2,2,2-trifluoroethanone (1 equivalent of diketone, 1.2 equivalents of $ZnEt_2$, 0.1 equivalents of $\mathbf{L^1}$), which bears both methyl- and trifluoromethylketone groups (See the Supporting Information). On the other hand, long reaction times needed for the addition reactions with $\mathbf{L^2}$ (1 equivalent of diketone, 3 equivalents of $ZnEt_2$, 0.1 equivalents of $\mathbf{L^2}$), the large excess of $ZnEt_2$, and the secondary reduction reaction of trifluoromethylketone group, impair the selectivity of this ligand towards the addition to methylketones.

The selective reactivity of $\mathbf{L^1}$ and $\mathbf{L^2}$ can be easily understood. Trifluoromethylketones show a higher reactivity towards nucleophilic addition but a lower ability for the coordination to Lewis acids,[22] which impairs the operation of a Lewis acid/Lewis base assisted mechanism (Scheme 1). Moreover, ^{13}C NMR data and theoretical calculations[23] support that the electronic density at the ketone carbon increases when the methyl group is replaced by the strongly electron withdrawing trifluoromethyl group. This counterintuitive effect would predict the opposite reactivity trend, i.e. a slower addition for the fluorinated ketone, but the difference in terms of reactivity is successfully explained when taking into account the higher energy stabilization of the LUMO in trifluoromethyl ketones compared to methylketones.[23] From a steric point of view, the trifluoromethyl group has a van der Waals molar volume similar to the isopropyl group,[24] which disfavors the formation of the highly sterically hindered TS proposed by Ishihara (Scheme 1).

However the catalytic enantioselective alkylations of trifluoromethylketones described here fit with the mechanism proposed by Hevia (Scheme 2). The step where the enantioselectivity is induced should be the formation of the nucleophilic addition product. The departure of $\mathbf{L^1}$ from the crowded alkyl(alkoxide) zinc intermediate is probably easy and allows for the coordination of $\mathbf{L^1}$ to free ZnR_2 and regeneration of the alkylating reagent [$ZnR_2(\mathbf{L^1})$].

In view of the moderate results with $\mathbf{L^2}$, we focused on $\mathbf{L^1}$ to extend the addition of dimethyl- and diethylzinc to new commercially available trifluoroacetophenone derivatives containing EWG and EDG substituents (Scheme 4). The results of the enantioselective alkyl addition, using ZnR_2 and the chiral diamine $\mathbf{L^1}$ ($ZnMe_2$:trifluoromethyl ketone:$\mathbf{L^1}$ = 1.2:1:0.1), are summarized in Table 2.

$ZnMe_2$ gave, after hydrolysis, the corresponding 2-aryl-1,1,1-trifluoropropan-2-ols (Table 2), usually in high yields, with high to moderate ee. The ketones **12** (entry 7) and **17** (entry 12) were significantly less reactive, giving moderate yields, and for **12a** the ee was only 10%. In the case of the methoxyketones (entries 11 and 12) the increased steric hindrance close to the carbonyl group in **17**, as compared to **16**, affects negatively the yield but not the ee.

^{t}BuBOX and $\mathbf{L^3}$ (Figure 1) gave also high yields (90-99%) in the reaction with **1** or **9**, but the ee values were very low (^{t}BuBOX: 5% for **1a**, 7% for **9a**; $\mathbf{L^3}$: 15% for **1a**[18]). The nonchiral TMEDA was also an efficient catalyst, but only for racemic

Table 2. Addition of $ZnMe_2$ or $ZnEt_2$ to aryl 2,2,2-trifluoroacetophenones using $\mathbf{L^1}$

		$ZnMe_2$				**$ZnEt_2$**				
				addition				addition		red.
entry	ket	t (h)	T (ºC)	yield (%)	ee (%)	t (h)	T (ºC)	yield (%)	ee (%)	yield (%)
1	**1**	24	–37/rt	98[(a)]	83[a]	15	–37	92[a]	84[a]	5[a]
						48	–60	97[a]	92[a]	0[a]
2	**7**	48	–37/rt	100	75	48	–40	97	86	1
3	**8**	24	–37/rt	95[a]	76[a]	1	–40	96	55	0
		48	–65	95[a]	81[a]	48	–40	98	51	0
4	**9**	24	–37/rt	95[a]	70[a]	1	–40	96	64	0
5	**10**	72	–37/rt	97[a]	54[a]	48	–60	75	89	0
6	**11**	48	–37/rt	97[a]	64[a]	48	–40	56	72	1
7	**12**	72	–37/rt	65	10	48	–40	8	38	5
8	**13**	72	–37/rt	99[a]	81[a]	48	–40	78	77	1
9	**14**	48	–37/rt	99	61	48	–40	38	76	1
10	**15**	48	–37/rt	99	58	48	–40	39	64	1
11	**16**	72	–37/rt	96	52	48	–40	7.5	49	0.5
12	**17**	48	–37/rt	59	55	72	–40	14	6	52
13	**18**	24	–37/rt	95[a]	60[a]	48	–40	37	30	0.9

[a] Previously reported data[18]

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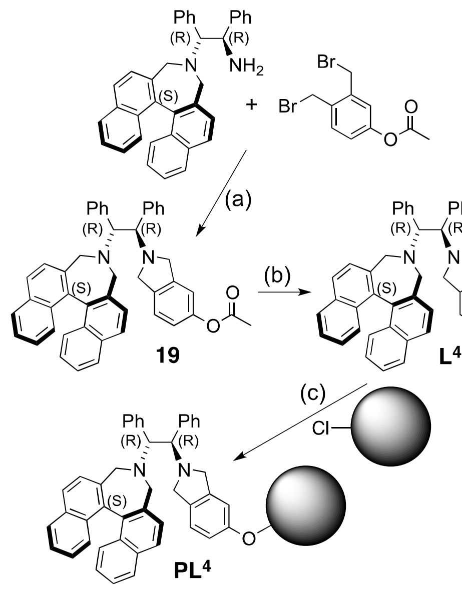

Scheme 5. (a) NEt_3, CH_2Cl_2, 40 ºC, 24 h. (b) $NaBO_3{\cdot}4H_2O$, MeOH, 45 ºC, 6 h. (c) Cs_2CO_3, DMF, 45 ºC, 3 days

syntheses, and the reaction hardly occurs in the absence of diamines. It is worth noting that good yield and enantioselectivity is obtained only for ligand **L¹**, with very bulk substituents on both nitrogen atoms.

In the asymmetric addition of $ZnEt_2$ the reduction products (**1c**, **7c**-**18c**) compete with the addition ones (**1b**, **7b**-**18b**). Table 2 collects the different yields and enantiomeric excesses, obtained by ^{19}F NMR integration of the reaction mixtures after hydrolysis, and by GC or HPLC analysis. When the reaction was not complete we detected in some cases, the hydration diol $RC(OH)_2(CF_3)$ together with the starting ketone. The undesired reduction products were usually very little or not observed.

Again, the phenyl substituents 4-^{i}Pr and 2-OMe entries (7 and 12) seem to influence negatively the yield and the ee of the addition products and, in the case of 2-OMe, a large amount of the reduction product (52%) is obtained. The observed reduction products are always racemic, which is relevant considering that, in the absence of any ligand, $ZnEt_2$ reacts slowly with trifluoroacetophenone in the same conditions giving almost exclusively the reduction product (82%). Compared to results reported for ^{t}BuBOX as a ligand,17 the reactions with **L¹** are slower, but the ee's obtained are higher. The yields in addition products and the ee values obtained for $ArCOCF_3$ ketones in Table 2 show a moderate sensitivity of the reaction to the Ar group substituent. Electron-withdrawing substituents directly linked to the aromatic ring should lower the LUMO energy level, favoring the nucleophilic attack and increasing the turnover frequencies of the catalyst. In effect, high yields (96 %) are obtained in 1 h for Cl and Br derivatives. As a general rule, ee's are lower for $ZnEt_2$ than for the smaller $ZnMe_2$, and increase with the halogen electronegativity 49% (Br) < 51% (Cl) < 86% (F). On the other hand, substrates bearing electron-donating substituents, namely MeO–, MeS–, EtS–, $MeCO_2$–, show lower yields but similar ee's compared to the reactions with $ZnMe_2$. The bulkiness of the 4-alkyl group in the aryl (entries 5-7) seems to have a bigger impact on the addition yields for $ZnEt_2$ (92% (H) < 75% (Me) < 56% (Et) < 8% (^{i}Pr) than for $ZnMe_2$. This trend can also be explained attending at the inductive effect of these groups.

Table 3. Successive catalytic cycles in the addition of $ZnEt_2$ to trifluoroacetophenone with **PL⁴**. Elemental analysis of the resin after use is provided.[a]

run	addition product yield (%)	addition product ee (%)	add/red	C/H/N (%)[b]	Unassigned (%)[c]
1	92	57	11.5	78.66/6.49/1.68	13.17
2	90	58	9.0	67.96/6.09/1.52	24.43
3	82	57	4.6	62.54/5.43/1.30	30.73
4	66	51	1.5	58.79/5.13/1.23	34.85
5	69	57	1.6	55.81/4.97/1.15	38.07

[a)] Experimental C/H/N (%) of the resin used in the first catalytic run: 88.93/6.81/1.70. [b)] Calculated C, H, N (%) considering 100% of functionalization: 90.09/6.95/1.88. [c)] "Unassigned"= 100- (%C + %H + %N); 100 % functionalized resin has a theoretical value of 1.08%.

As observed along the discussion so far, the bulkiness of the substituents on both nitrogen atoms has a capital influence on the catalytic performance. On the other hand, these encumbered ligands are hard and expensive to produce. This encouraged us to design a heterogeneous catalyst by grafting a suitable ligand into a polymer backbone that might be considered the second bulky substituent. Catalyst immobilization offers additional advantages such as an easier handling, separation and isolation, and the possibility of recycling the expensive N–N' catalyst

Many different supports have been used for immobilization,[25] including soluble and insoluble polymers, silica gel, and dendrimers. Merrifield resins have been successfully used in the addition reaction of diethylzinc to aldehydes.[26] A suitable ligand should have an anchoring functionality to react with the chloro substituents of the resin. In order to try to emulate the good performance of **L¹**, a molecular modification in the diamine was needed to introduce an anchoring point far away from the coordinating sites of the catalyst (the N atoms). However, our attempts to introduce an appropriate functionality on **L¹** were cumbersome and failed. Fortunately, it was possible to obtain a **L³** derivative bearing a hydroxy group in position 5 of the isoindoline group. The synthesis of **L⁴**, is shown in Scheme 5.

Derivatization of a commercially available Merrifield resin (1.1 mmol Cl/g resin) with **L⁴** was achieved by treatment with Cs_2CO_3 in DMF, and afforded the polymer ligand **PL⁴**. The level of functionalization was estimated by elemental analysis (see the Experimental Section).

The catalytic activity of the functionalized resin for the addition of $ZnEt_2$ to trifluoroacetophenone was checked. Compared to its homogeneous counterpart **L³**, the supported catalyst displays similar yields (**L³**: 95%; **PL⁴**: 92%) but lower e.e. (**L³**: 76%; **PL⁴**: 57%). This behavior is quite frequent for heterogeneized catalysts, as the supported catalyst becomes more sterically hindered, impairing the approach of substrate to the active sites and modifying the transition states involved in the catalysis.

The recyclability of the functionalized resin was checked in five successive catalytic runs of addition of $ZnEt_2$ to trifluoroacetophenone (Table 3). According to these results, this novel heterogenized catalyst maintains its stereoselectivity untouched after 5 recycling runs. The yields are very similar in the first two runs and then fall from excellent to fairly good. However, the addition/reduction ratio steadily diminishes after each run and becomes severe after the third recycling. Elemental microanalysis of the resins, before and after every catalytic cycle, clearly show that resin aging due to the recycling of the catalyst produces a progressive decrease in the C,H,N percentage. The accumulation of hydrolyzed zincate species in the polymer matrix, confirmed by ICP-MS, would explain the marked loss of activity after two catalytic cycles. Despite its well-known aqueous chemistry, attempts to leach Zn(II) species using acidic or basic aqueous solutions were unsuccessful. These zincate species may be coordinated with the ligand and their presence may also modify the swelling ability of the resin.

Conclusions

The selective reactivity of the ligands used in the addition of ZnR_2 (R = Me, Et) to methyl- and trifluoromethyl-ketones offers an enantioselective access either to chiral non fluorinated or to chiral fluorinated tertiary alcohols, without the assistance of protecting groups. Focusing on the later we have widened the scope of the addition reaction of $ZnEt_2$ to trifluoromethylketones. In this sense, we have used two different approaches to obtain the presence of the chiral trifluoromethyl carbinol fragment: the homogeneous catalyst $\mathbf{L^1}$ provides excellent yields and very high ee whilst immobilized tail-tied ligand $\mathbf{PL^4}$ affords moderate ee but allows an easier handling and recycling.

Experimental Section

General Information. All reactions were performed under argon atmosphere. Toluene was dried over sodium and distilled. Flash chromatography was performed on silica gel (silica gel 60, 230-400 mesh, Merck). Commercial products were purchased from Aldrich Chemical Co. and Fluorochem and used without further purification. Analytical gas chromatography was performed on a Hewlett-Packard 5890 Series II machine equipped with a LIPODEX-E (25 m, 0.25 mm ID) capillary column. NMR experiments were performed on a Bruker AV400 spectrometer (400.13 MHz for ^{1}H, 100.61 MHz for ^{13}C, 376.50 MHz for ^{19}F, and 161.96 MHz for ^{31}P). Chemical shifts (δ) are given in ppm downfield from Me_4Si and are referenced as internal standard to the residual solvent (unless indicated) $CDCl_3$ (δ 7.26 for ^{1}H and δ 77.0 for ^{13}C). H_3PO_4 was used as external standard (δ 0.0 for ^{31}P). Coupling constants, J, are reported in hertz (Hz). TLC was carried out on SiO_2 (silica gel 60 F254, Merck), and the spots were located under UV light. Optical rotation was measured on a Perkin-Elmer 343 apparatus. Mass spectra were recorded on an Agilent Technologies 5973 Network apparatus. High resolution mass spectra were recorded on an Bruker ESI-QTOF Maxis Impact apparatus.

Preparation of $\mathbf{L^2}$. NaH (51 mg, 2.14 mmol) was added under argon to a solution of *2-((1R,2R)-2-((S)-3H-dinaphtho[2,1-c:1',2'-e]azepin-4(5H)-yl)-1,2-diphenylethyl)isoindolin-5-yl acetate*[27] (500 mg, 1.02 mmol) in anhydrous DMF (30 mL) at –20 °C. The solution was stirred for 60 minutes while the temperature increases and di(naphthalen-1-yl)phosphinic chloride (378 mg, 1.12 mmol) was added at –10 °C. After the reaction was complete (3 days), it was quenched with saturated ammonium chloride solution, extracted with dichloromethane, and dried over $MgSO_4$. The solvent was removed and the crude product was purified by column chromatography (1/1 hexanes/ethyl acetate). ^{1}H NMR (400.13 MHz, $CDCl_3$), δ (ppm): 3.57 (dd, J = 236.37, 12.13 Hz, 4H, $2CH_2$); 3.95 (d, J = 7.20 Hz, 1H, CH) 4.98 (br, 1H, NH); 5.28 (br, 1H, CH); 6.86 – 6.92 (m, 6H, Ar); 7.06 – 7.016 (m, 8H, Ar); 7.18 (q, J = 7.90 Hz, 2H, Ar); 7.23 – 7.28 (m, 2H, Ar); 7.31 – 7.39 (m, 2H, Ar); 7.41 (d, J = 8.34 Hz, 2 H, Ar); 7.46 – 7.50 (m, 2H, Ar); 7.61 – 7.66 (m, 4H, Ar); 7.72 – 7.74 (m, 2H, Ar); 7.85 – 7.93 (m, 4H, Ar); 8.71 (t, J = 8.34 Hz, 2H, Ar). ^{13}C NMR (100.61 MHz, $CDCl_3$), δ (ppm): 52.64 (s, 2C, $2CH_2$); 54.72 (s, 2C, 2CH); Ar: 124.17, 124.24, 124.32, 124.39, 125.35, 125.56, 125.86, 126.15, 126.62, 126.74, 126.78, 126.93, 127.28, 127.50, 127.55, 127.60, 127.65, 127.98, 128.24, 128.32, 128.49, 128.50, 128.55, 128.82, 130.05, 131.14, 132.60, 132.65, 132.58, 132.70, 132.85, 132.94, 133.39, 133.43, 133.52, 133.62, 133.65, 133.75, 134.22, 134.32, 134.67, 136.81, 139.07. ^{31}P NMR (161.96 MHz, $CDCl_3$), δ (ppm): 29.81 (s). HRMS: Calculated mass for $C_{56}H_{43}N_2OP$: 790.318577; Measured mass: [M+H] 791.318201. Chemical yield: 42%.

Preparation of 19. Et_3N (0.09 mL, 0.612 mmol) was added to a solution of *2-((1R,2R)-2-((S)-3H-dinaphtho[2,1-c:1',2'-e]azepin-4(5H)-yl)-1,2-diphenylethyl)isoindolin-5-yl acetate* (150 mg, 0.306 mmol) and 3,4-bis(bromomethyl)phenyl acetate (98 mg, 0.306 mmol) in anhydrous CH_2Cl_2 (6 mL) under argon and the mixture stirred for 24 hours at 40 °C. The solvent was removed and the crude product was purified by column chromatography (4/1 hexanes/ethyl acetate). Rf 0.16 (4/1 hexanes/ethyl acetate). ^{1}H NMR (400.13 MHz, $CDCl_3$), δ (ppm): 2.26 (s, 3H, CH_3); 3.74 (dd, J = 348.54, 12.17, 4H, $2CH_2$); 3.80 (dd, J = 12.20, 10.87, 2H, CH_2); 4.04 (dd, J = 12.80, 6.51, 2H, CH_2); 4.34 (dd, J = 83.28, 4.98, 2H, 2CH); 6.78-6.83 (m, 2H, Ar); 7.00-7.26 (m, 13H, Ar); 7.32 (d, J = 8.31, 2H, Ar); 7.43-7.48 (m, 4H, Ar); 7.87 (d, J = 8.30, 2H, Ar); 7.94 (d, J = 8.09, 2H, Ar). HRMS: Calculated mass for $C_{46}H_{38}N_2O_2$: 651.3012; Measured mass: 651.3002. Chemical yield: 72.4%.

Preparation of $\mathbf{L^4}$. Compound **19** (90.9 mg, 0.14 mmol) and $NaBO_3.4H_2O$ (21 mg, 0.14 mmol) were dissolved in methanol and the mixture was stirred for 6 hours at 40 °C. The solvent was removed and the crude product was purified by column chromatography (1/1 hexanes/ethyl acetate). Rf 0.24 (1/1 hexanes/ethyl acetate). ^{1}H NMR (400.13 MHz, $CDCl_3$), δ (ppm): 3.79 (dd, J = 333.85 Hz, 11.65 Hz, 4H, $2CH_2$); 3.72 (dd, J = 11.65, 7.76 Hz, 2H, CH_2); 4.04 (dd, J = 17.86, 12.42 Hz, 2H, CH_2); 4.41 (dd, J = 43.48, 6.21 Hz, 2H, 2CH); 6.35 (br, 1H, Ar); 6.50 (dd, J = 8.05, 2.26 Hz, 1H, Ar); 6.85 (d, J = 8.09 Hz, 1H, Ar); 7.04-7.27 (m, 14H, Ar); 7.43-7.48 (m, 4H, Ar); 7.83 (d, J = 8.31 Hz, 2H, Ar); 7.92 (d, J = 8.06 Hz, 2H, Ar). ^{13}C NMR (100.61 MHz, $CDCl_3$), δ (ppm): 53.36, 58.42, 59.07, 60.47, 70.35, 71.00, 109.51, 113.99, 122.81, 125.28, 125.53, 126.83, 126.98, 127.36, 127.40, 127.54, 127.84, 128.16, 128.22, 130.11, 130.22, 131.21, 132.93, 134.12, 134.96, 139.02, 139.21, 141.22, 155.02. Chemical yield: 82%.

Preparation of $\mathbf{PL^4}$. A mixture of a Merrifield resin (164 mg, 0.181 mmol Cl) and Cs_2CO_3 (241 mg, 0.235 mmol) in anhydrous DMF (3 mL) was shaken for 45 minutes under argon atmosphere. A solution of $\mathbf{L^4}$ (143 mg, 0.235 mmol) in anhydrous DMF (3 mL) under argon, was then added. The mixture was shaken for 24 hours. The resin was filtered and washed sequentially with: DMF; H_2O/DMF; H_2O; $Na_2CO_3/NaHCO_3$ solution (pH 9); H_2O; MeOH; Toluene; CH_2Cl_2. The resin was dried in vacuo. Chemical yield was determined by elemental analysis: 90%. Analysis calculated for $C_{111.7}H_{102.7}N_2O$: C, 90.09; H, 6.95; N, 1.88. Found: C, 88.93; H, 6.81; N, 1.70

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General Procedure for the Enantioselective addition to methyl ketones. Diethylzinc (1.1 M in toluene, 0.68 mL, 0.75 mmol) was added to a solution of diamine L^2 (19.8 mg, 0.025 mmol, 10 mol %) in anhydrous toluene (2 mL) under argon at –35 °C. The solution was stirred for 30 minutes while the temperature increases. Methyl ketone (0.25 mmol) was added at –65 °C and the solution stirred for four days while the temperature go up. After completion, the reaction was quenched with saturated ammonium chloride solution, extracted with ether, and filtered over silica gel.

General Procedure for the Enantioselective addition to trifluoromethyl ketones. Diethylzinc or dimethylzinc (1.0 M in heptane, 0.58 mL, 0.58 mmol) was added to a solution of diamine L^1 (37 mg, 0.048 mmol, 10 mol %) in anhydrous toluene (0.6 mL) under argon at –20 °C. The solution was stirred for 30 minutes while the temperature goes up. Trifluoromethyl ketone (0.48 mmol) was added at –40 °C and this temperature is retained. After the reaction was complete, it was quenched with saturated ammonium chloride solution, extracted with ether, and filtered over silica gel. The solvents were removed under reduced pressure.

Characterization of methyl and trifluomethylethyl alcohols. Characterization of literature methyl and trifluoromethyl alcohols were made according to spectroscopic data described elsewhere. Full spectroscopic data for new compounds are described below:

1,1,1-trifluoro-2-(4-isopropylphenyl)butan-2-ol (12b). The ee was determined by GC on LIPODEX E using ethanol as the mobile phase, at 110 °C isothermal, t_1 = 21.1 min, t_2 = 27.7 min. 38% ee. Chemical yield was determined by ^{19}F NMR (376.50 MHz, $CDCl_3$) δ = –80.49, 8%. ^{1}H NMR (400.13 MHz, $CDCl_3$) δ = 0.81 (t, J = 7.5 Hz, 3H), 2.04 (m, 1H), 2.22 (m, 1H), 2.27 (s, 1H), 1.25, 1.27 (d, J = 7.23 Hz, 6H), 7.27 (m, 2H), 7.43 (m, 2H). MS: Calculated mass for $C_{13}H_{17}F_3O$: 246.12; Measured mass: 246.08.

2-(3,5-dimethylphenyl)-1,1,1-trifluorobutan-2-ol (13b). The ee was determined by GC on LIPODEX E using ethanol as the mobile phase, at 100 °C , isothermal, t_1 = 27.8 min, t_2 = 29.4 min. 77% ee. Chemical yield was determined by ^{19}F NMR (376.50 MHz, $CDCl_3$) δ = –80.35, 78%. ^{1}H NMR (400.13 MHz, $CDCl_3$) δ = 0.81 (t, J = 7.4 Hz, 3H), 2.03 (m, 1H), 2.20 (m, 1H), 2.23 (s, 1H), 2.35 (s, 6H), 6.99 (m, 1H), 7.13 (m, 2H). MS: Calculated mass for $C_{12}H_{15}F_3O$: 232.11; Measured mass: 232.10.

2-(4-(ethylthio)phenyl)-1,1,1-trifluorobutan-2-ol (15b). The ee was determined by HPLC on Chiralpak AD using n-hexane/IPA 95:5 as the mobile phase, t_1 = 15.7 min, t_2 = 19.8 min. 64% ee. Chemical yield was determined by ^{19}F NMR (376.50 MHz, $CDCl_3$) δ = –80.60, 39%. ^{1}H NMR (400.13 MHz, $CDCl_3$) δ = 0.80 (t, J = 7.5 Hz, 3H), 1.34 (t, J = 7.3 Hz, 3H), 2.04 (m, 1H), 2.20 (m, 1H), 2.29 (s, 1H), 2.97 (q, J = 7.3 Hz, 2H), 7.32 (m, 2H), 7.42 (m, 2H). MS: Calculated mass for $C_{12}H_{15}F_3OS$: 264.08; Measured mass: 264.07.

1,1,1-trifluoro-2-(2-methoxyphenyl)butan-2-ol (17b). The ee was determined by GC on LIPODEX E using ethanol as the mobile phase, 20 min at 110 °C, then 30 min at 130 °C, ramp 10 °C /min, t_1 = 17.5 min, t_2 = 18.2 min. 20% ee. Chemical yield was determined by ^{19}F NMR (376.50 MHz, $CDCl_3$) δ = –80.05, 17%. 1H NMR (400.13 MHz, $CDCl_3$) δ = 0.92 (t, J = 7.3 Hz, 3H), 2.05 (m, 1H), 2.22 (m, 1H), 3.94 (s, 3H), 6.24 (s, 1H), 7.00-7.05 (m, 2H), 7.24 (bs, 1H), 7.33-7.37 (m, 1H). MS: Calculated mass for $C_{11}H_{13}F_3O_2$: 234.09; Measured mass: 238.08.

Methyl-4-(1,1,1-trifluoro-2-hydroxybutan-2-yl)benzoate (18b). The ee was determined by HPLC on Chiralpak AD using n-hexane/IPA 95:5 as the mobile phase, t_1 = 5.6 min, t_2 = 6.7 min. 30% ee. Chemical yield was determined by ^{19}F NMR (376.50 MHz, $CDCl_3$) δ = –79.77, 98%. ^{1}H NMR (400.13 MHz, $CDCl_3$) δ = 0.78 (t, J = 7.46 Hz, 3H), 2.07 (m, 1H), 2.24 (m, 1H), 2.39 (s, 1H), 3.92 (s, 3H), 7.64 (m, 2H), 8.06 (m, 2H). MS: Calculated mass for $C_{12}H_{13}F_3O_3$: 262.08; Measured mass: 262.05.

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Notes and references

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